# Imon Kalyan Ghosh

imonghose@gmail.com +4915771353412 Magdeburg, Germany LinkedIn Github

## PROFILE

- Master's student in Data Science specializing in generative AI, with 3+ years of professional experience in software development, API integration and databases
- Skilled in workflow automation, building LLM-powered applications, design and orchestration of multi-agent workflows and RAG pipelines, deployment and monitoring of AI systems
- Strong communication skills with structured and independent way of working

## EDUCATION

**Master of Science In Data and Knowledge Engineering,** Apr 2024 – present
*Otto-von-Guericke-University Magdeburg, Current GPA: 1.5 (German scale)* Magdeburg, Germany

**Bachelor of Technology in Information Technology,** Aug 2016 – Jul 2020
*Techno India Saltlake, CGPA: 8.35/10* Kolkata, India

## SKILLS SUMMARY

- **Languages:** Python, R, SQL, Java, Kotlin, JavaScript, TypeScript, Dart
- **Deep Learning Frameworks:** PyTorch, TensorFlow
- **LLM Orchestration:** LangChain, LangGraph, Crew AI, Autogen, PydanticAI, Agents SDK
- **LLM Tools:** MCP, OpenAI API, Hugging Face
- **Automation Tools:** n8n
- **Backend Frameworks:** Fast API, Spring
- **UI Frameworks:** React, Next.js, Flutter, Streamlit, Gradio
- **Databases:** Relational – (Oracle DB, MySQL, PostgreSQL) , Vector – (pgvector, Chroma, Qdrant)
- **Data Analysis:** NumPy, Pandas, Matplotlib, Plotly, ggplot2
- **Observability:** Langfuse, LangSmith, AWS CloudWatch
- **Project Tools :** Jira, Notion, ClickUp, Confluence
- **Cloud & DevOps:** Git, Modal, Vercel, Docker, AWS

## PROJECTS

**Multimodal Agentic RAG SaaS Application**– (Link) Jan 2026
- Built a RAG application (FastAPI + Next.js) for querying over URLs, multiple file types, audio, and embedded images
- Ingestion pipeline with Crawl4AI for intelligent web-crawling and Docling for document parsing and hybrid chunking
- Designed a structured knowledge storage using PostgreSQL and pgvector for embeddings, metadata, and semantic search
- Implemented a PydanticAI RAG agent with tool routing to enable adaptive and hybrid retrieval strategies
- RAG evaluation using DeepEval (>90% on all critical metrics), observability using Langfuse and AWS Cloudwatch
- Containerized via Docker, pushed to Amazon ECR and deployed to AWS AppRunner

**Automated Multi-Agent Trading Simulation Platform** – (Link) Dec 2025
- Engineered a fully automated multi-agent trading simulation platform where a researcher ingests real-time market data and Trader agents run autonomous execution cycles with configurable scheduling and dynamic strategy optimization
- Implemented an MCP-integrated toolchain to standardize agent tool access and streamline orchestration
- Developed a simulated brokerage backend supporting end-to-end trading and account management, persisted in SQLite
- Delivered end-to-end observability via a real-time trading floor that streams custom agent logs, critical metrics, and interactive charts for monitoring performance and debugging

**Job search automation pipeline using n8n -** (Link) Sept 2025
- Built a scalable low-code job-search automation workflow using n8n and Bolt, integrating LinkedIn data ingestion, AI-based relevance scoring, and automated resume tailoring
- Utilised Apify and Google API to scrape and log structured job data to a personal Google Sheets repository
- Implemented a chatbot interface serving as a webhook trigger, which captures job url input and outputs tabular results
- Demonstrated effective orchestration of AI services, external APIs, and low-code platforms

**AI - powered Deal Discovery and Pricing Platform** – (Link) Aug 2025

- Architected a LLM and RAG-based multi-agent system for real-time deal discovery and price estimation
- Built a scanning agent to normalize heterogeneous RSS feeds via LLM parsing and stream deals to a real-time dashboard
- Combined a Frontier Agent backed by a 400k-sample Chroma vector store and a Specialist Agent powered by a custom fine-tuned Llama model, enabling robust price prediction
- Orchestrated planning, scoring, and alerting to rank best deals by estimated discount and trigger real-time notifications

**Natural Language to SQL Pipeline** – (Link) Jun 2025

- Developed a natural-language-to-SQL assistant using LangChain and Streamlit enabling users to query relational datasets
- Architected an end-to-end workflow with conversation memory, guardrails, dynamic few shot guidance and table-aware prompting to ensure contextual, read-only, schema consistent SQL generation and data retrieval
- Designed a metadata-driven, modular pipeline that generalizes across database schemas
- Added LangSmith observability to enable effective monitoring, debugging, and quality evaluation

## PROFESSIONAL EXPERIENCE

**STUDENT DEVELOPER ,** *Otto Von Guericke University* June 2025 – Current | Magdeburg, Germany

- Developed the OVGU university application using Spring, Vue and Flutter, enhancing usability and student experience
- Created and maintained Dockerized services on Linux-based VMs for development and production environments
- Managed Android and iOS releases of the Flutter app, including packaging, app store submission, and updates
- Handled end-to-end deployment, monitoring, and incident response to ensure high availability and reliable operation

**RESEARCH ASSISTANT,** *Leibniz Institute For Resilience Research* April 2025 – Dec 2025 | Mainz, Germany

- Built a Python signal-processing pipeline to ingest, QC, and extract ECG/EMG features from physiological data streams
- Performed data preprocessing and behavioral statistical analysis in R using ggplot2; delivered graphical reports

**SOFTWARE ENGINEER,** *Nomura Research Institute Financial Technologies India Pvt. Ltd.* Aug 2020 – Feb 2024 | Kolkata, India

- Developed critical features for a multi-tenant trade back-office system for the Australian Stock Exchange
- Developed real-time trading dashboards and UI modules using JSP, JavaScript and Spring Boot
- Implemented end-to-end business workflows using Spring, ActiveMQ and Oracle DB
- Created technical documentation and impact analysis reports in Confluence
- Fixed high priority bugs and enhanced overall system reliability

## LANGUAGES

**English** (C1) | **German** (A2) | **Bengali** (Native)

## CERTIFICATES

- LLM Engineering certification from Udemy
- Agentic AI certification from Udemy